Product Requirements Document (PRD)

Product: Internship + Quizathon + Projectathon Platform (working name: *InternSetu*)
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1. Name Suggestions

Orbita Tech - Your Career, In Orbit
Arohan Technologies - Rise With Skills
Skillforge Technologies **-Where Skills Are Forged**

Recommendation: InternSetu (primary), SkillNest (backup). Check domain availability (.in / .com) and trademark before finalising.

2. Vision and Goals

Vision: One platform where any student, from any stream, can do a virtual internship, test knowledge in quizzes, and prove skills in coding challenges, and get a QR-verifiable certificate.

Goals

- Zero manual work: registration to certificate is fully automated.
- Support 50-60 domains across Engineering, MBA, Electronics, Electrical, Mechanical, Computer Basics, Excel, DevOps and more.
- Every certificate is verifiable by QR + unique ID.
- Generate revenue through certificate, quiz and Projectathon fees.

3. Target Users

- **Student:** college students (BBA, MBA, BCA, B.Tech, Diploma etc.) who want internship experience and certificates.
- **Admin:** your team; manages domains, tasks, templates, quizzes, payments.
- **Verifier (public):** recruiters, colleges, anyone verifying a certificate.

4. Product Scope: 3 Modules

1. Internship Portal
2. Quizathon
3. Projectathon

Shared: authentication, payments, certificate engine, verification page, admin panel.

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5. Module 1: Internship Portal

5.1 User Flow

1. Student registers (name, email, phone, college, course, year, password / Google login).
2. Selects **one domain** from the 50-60 list and picks duration and start date.
3. System **auto-creates the dashboard** in the student's name and generates a **Unique Intern ID** (example: `ISX-2026-ENG-004821`).
4. Dashboard opens with a prominent **Download Your Offer Letter** button.
5. Offer letter is generated from the admin-uploaded template with: Name, Domain, Duration, Start Date, End Date, Unique ID, Issue Date.
6. After download, **Joining Task 1** appears: post offer letter on LinkedIn and submit the post link.
7. On valid submission, **4-5 domain-specific tasks unlock**.
8. Each task has input fields for **GitHub link** and **LinkedIn post link** (and an optional notes field).
9. When **50% of tasks are completed**, the **Certificate** section unlocks.
10. Student clicks Unlock Certificate and sees a **charges breakdown**; pays online.
11. After successful payment, certificate is **auto-generated** (PDF) with a QR code.
12. Student can download the certificate and share it on LinkedIn.

5.2 Functional Requirements

ID	Requirement	Priority
INT-01	Registration, login, forgot password, email verification	Must

ID	Requirement	Priority
INT-02	Domain catalogue (50-60 domains) with description, tasks and category filter	Must
INT-03	Auto dashboard creation after registration/domain selection	Must
INT-04	Unique Intern ID generation (never repeats)	Must
INT-05	Offer letter generation from template (placeholders filled automatically)	Must
INT-06	Joining Task 1 (LinkedIn post link) shown after offer letter download	Must
INT-07	Link validation (must be a valid linkedin.com URL; GitHub tasks must be github.com URL)	Must
INT-08	4-5 tasks per domain unlock after Joining Task approval	Must
INT-09	Task submission with GitHub + LinkedIn link inputs; status: Pending / Submitted / Approved / Rejected	Must
INT-10	Progress bar showing % tasks completed	Must
INT-11	Certificate unlock at 50% task completion and internship end condition	Must
INT-12	Charges breakdown page, online payment, receipt	Must
INT-13	Auto certificate PDF generation with QR code and Unique ID	Must
INT-14	Admin review of submissions (manual or auto-approve mode)	Should
INT-15	Email/WhatsApp notifications (task unlocked, certificate ready)	Should
INT-16	Re-download of offer letter and certificate anytime	Must

5.3 Domain Examples (50-60 total)

- **Engineering/Tech:** Python, Java, Web Development, Data Science, AI/ML, Cyber Security, Cloud, DevOps, Android, UI/UX, Basic Computer, Excel, Power BI
- **Electronics/Electrical:** Embedded Systems, IoT, VLSI Basics, PCB Design, Power Systems, Electrical Design (AutoCAD)
- **Mechanical/Civil:** CAD/CAM, SolidWorks, Thermodynamics Projects, Robotics, Structural Basics
- **Management/MBA:** HR, Digital Marketing, Finance, Marketing, Operations, Business Analytics, Sales

(Final list to be decided by admin; the system must allow adding or removing domains without code changes.)

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6. Module 2: Quizathon

6.1 Flow

1. Student opens **Quizathon** and sees the list of active quizzes (20-25 domains running at any time, selected from the 60).
2. Selects a domain/quiz and sees: number of questions (10-15 MCQs), time limit, passing criteria (75%), **fee**.
3. Pays the fee; quiz starts immediately.
4. Timer-based MCQ test, one attempt per payment.
5. On submit, score shown instantly.
6. If score is **75% or above**, certificate unlocks (QR-verified, with Unique ID). Otherwise student can retry by paying again (admin configurable).

6.2 Functional Requirements

ID	Requirement	Priority
QZ-01	Admin can mark any domain's quiz Active/Inactive (rotation of 20-25 live quizzes)	Must
QZ-02	Question bank per domain; each attempt picks 10-15 random questions and shuffles options	Must

ID	Requirement	Priority
QZ-03	Fee display and payment before quiz start	Must
QZ-04	Timer, auto-submit on time-up, one question per screen or full page	Must
QZ-05	Anti-cheating basics: tab-switch warning, no back after submit, randomisation	Should
QZ-06	Pass mark 75%; certificate auto-generated with score and QR	Must
QZ-07	Attempt history and result page	Must
QZ-08	Bulk upload of MCQs via CSV/Excel	Must
QZ-09	Optional leaderboard per domain	Could

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7. Module 3: Projectathon

A hackathon-style coding event where students solve problem statements and submit projects.

7.1 Flow (proposed, to confirm)

1. Admin creates a Projectathon event: title, domains/tech tracks, problem statements, dates, fee (if any), rules, prizes.
2. Student registers (individual or team up to 4).
3. Problem statement(s) revealed at start time.
4. Student/team submits **GitHub repo link, demo/video link, short description** before the deadline.
5. Evaluation: admin/judges score on rubric (innovation, working demo, code quality, presentation). Optional auto-checks for coding questions.
6. Results and leaderboard published; **participation and winner certificates** (QR-verified) issued.

7.2 Functional Requirements

ID	Requirement	Priority
PJ-01	Event creation and management by admin	Must
PJ-02	Individual and team registration	Must
PJ-03	Problem statement release at scheduled time	Must
PJ-04	Submission form (GitHub, demo link, description) with deadline lock	Must
PJ-05	Judge panel login and scoring rubric	Should
PJ-06	Leaderboard and results page	Should
PJ-07	Participation / winner certificates with QR	Must
PJ-08	In-browser coding questions with auto-evaluation (phase 2)	Could

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8. Certificate Engine and Verification (Shared)

- Certificate types: Internship, Quiz, Projectathon (participation/winner).
- Each certificate contains: student name, domain/event, duration or score, issue date, **Unique ID**, **QR code**.
- QR opens: `https://<yourdomain>/verify?id=<UniqueID>`.
- **Verification page:** user scans the QR and enters the Unique ID; backend checks and shows student name, domain, dates/score, certificate type and status (Valid / Revoked). Invalid IDs show a clear "Not found" message.
- Admin can revoke a certificate.
- Certificates are stored so the student can always re-download them.

9. Payments

- Payment gateway such as Razorpay or Cashfree (UPI, cards, net banking).
- Charges are shown clearly **before** payment (base fee, GST, total).

- Fee configurable by admin per product (internship certificate, each quiz, Projectathon).
- Auto receipt/invoice by email; webhook-based confirmation so a certificate is never issued without confirmed payment.
- Refund and failed payment handling.

10. Admin Panel

- Dashboard: registrations, revenue, certificates issued.
- Domain management (add/edit/disable) and task management per domain.
- Upload offer letter and certificate templates with placeholders (`{{name}}` , `{{domain}}` , `{{start_date}}` , `{{end_date}}` , `{{duration}}` , `{{unique_id}}`).
- Review task submissions; approve/reject with remarks.
- Quiz and question bank management; activate/deactivate quizzes.
- Projectathon event and judge management.
- Student management (search, block, reset), certificate revoke, payment reports (export to Excel).

11. Student Dashboard Summary

- My Internship: domain, ID, dates, offer letter download, task list, progress, certificate.
- My Quizzes: attempts, scores, certificates.
- My Projectathons: registrations, submissions, results.
- Payments and receipts; profile settings.

12. Non-Functional Requirements

- **Performance:** pages load under 3 seconds; supports at least 5,000 concurrent users during quizzes.
- **Security:** HTTPS, hashed passwords, role-based access, rate limiting, input validation, protection against fake/duplicate submissions.
- **Mobile-first:** most students will use phones; fully responsive.
- **Reliability:** daily backups; 99.5% uptime target.
- **Scalability:** domains, questions and templates managed through admin without code changes.

- **Privacy:** student data protected as per Indian IT/DPDP rules; clear privacy policy and terms.

13. Suggested Tech Stack

- **Frontend:** Next.js (React) + Tailwind CSS
- **Backend:** Node.js (NestJS/Express) or Django
- **Database:** PostgreSQL (plus Redis for quiz sessions/timers)
- **File storage:** AWS S3 or Cloudflare R2
- **PDF/QR generation:** server-side (Puppeteer or pdf-lib, `qrcode` library)
- **Payments:** Razorpay / Cashfree
- **Email/SMS:** SES / Resend, WhatsApp API (optional)
- **Hosting:** Vercel + AWS/DigitalOcean

14. Core Data Entities

User, Domain, InternshipEnrollment, Task, TaskSubmission, OfferLetter, Quiz, Question, QuizAttempt, ProjectathonEvent, Team, ProjectSubmission, Certificate, Payment, AdminUser, Notification.

15. Release Plan

Phase	Scope	Timeline (suggested)
MVP (Phase 1)	Auth, 10-15 domains, dashboard, offer letter, tasks, payment, certificate with QR, verify page, basic admin	6-8 weeks
Phase 2	All 50-60 domains, Quizathon complete	4-5 weeks
Phase 3	Projectathon, judge panel, leaderboard, notifications	4-5 weeks
Phase 4	Auto-evaluated coding questions, analytics, referral program, mobile app	Ongoing

16. Success Metrics

- Registrations per month and registration-to-offer-letter download rate.
- Task completion rate; share of students reaching 50%.
- Certificate payment conversion rate and revenue per student.
- Quiz attempts and pass rate.
- Verification page visits (shows certificate credibility).

17. Risks and Mitigations

Risk	Mitigation
Fake or copied task submissions	Admin review, link validation, plagiarism spot checks
Quiz cheating	Randomised questions/options, timers, tab-switch warnings
Payment mismatch	Webhook confirmation before issuing certificate
Fake certificates	QR + Unique ID verification, revoke option
Peak load during quizzes	Caching, queueing, autoscaling
Content effort (60 domains x tasks x questions)	Start with MVP domains; bulk upload; reuse templates

18. Open Questions (to decide)

1. Exact fees for internship certificate, each quiz and Projectathon?
2. Internship durations offered (2 / 4 / 6 / 8 weeks)? Are start dates fixed batches or flexible?
3. Is the internship itself free and only the certificate paid?
4. Should tasks be auto-approved or manually reviewed?
5. Can a student take more than one domain at a time?
6. Quiz retry policy: unlimited paid retries or limited?
7. Projectathon: free or paid, individual or team, and prizes?

8. Do you need a college/campus-ambassador or referral module?
9. Legal: company registration, GST, terms and refund policy for the platform.